SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	Ragul.m	**Reg. No.:**	192524066
Section / Batch:	Slot A	**Date:**	24/07/2026

Code of Conduct

I RAGUL.M (192524066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: RAGUL.M

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

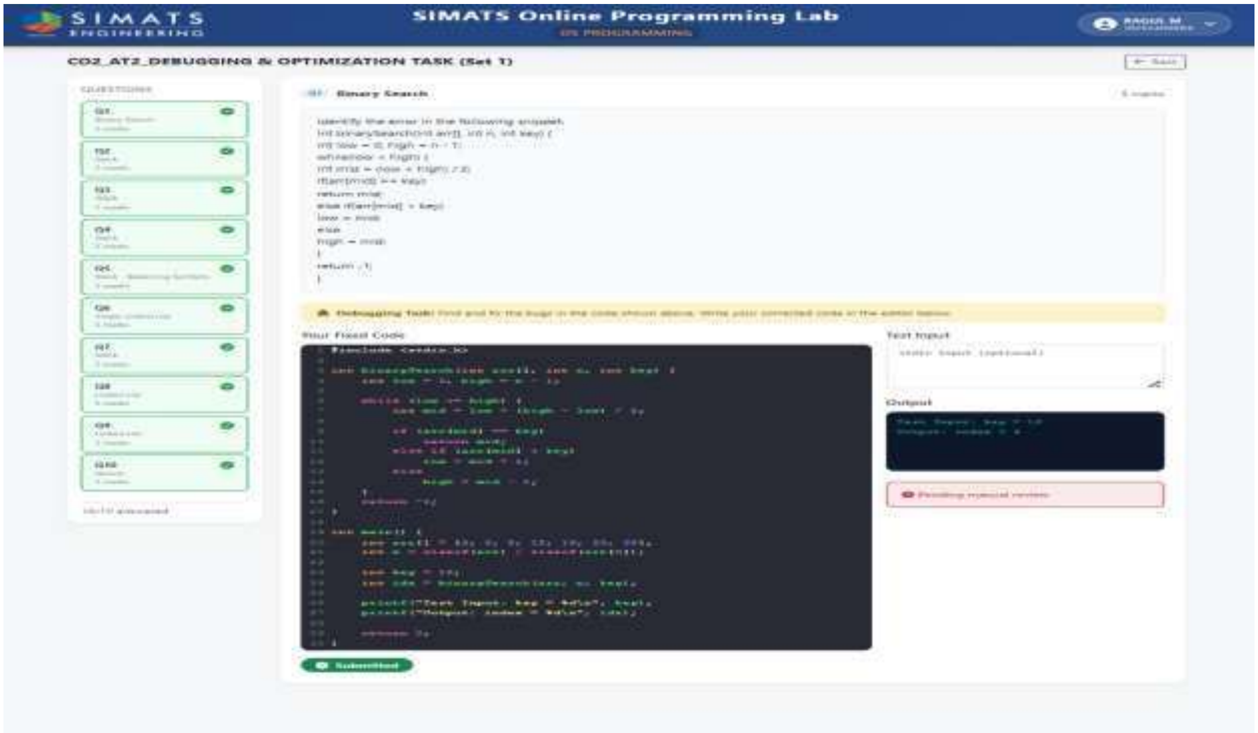
Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
	GRAND TOTAL:	100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edgecase errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors

Code Quality & Readability	20	Clean, modular, welldocumented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and welljustified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________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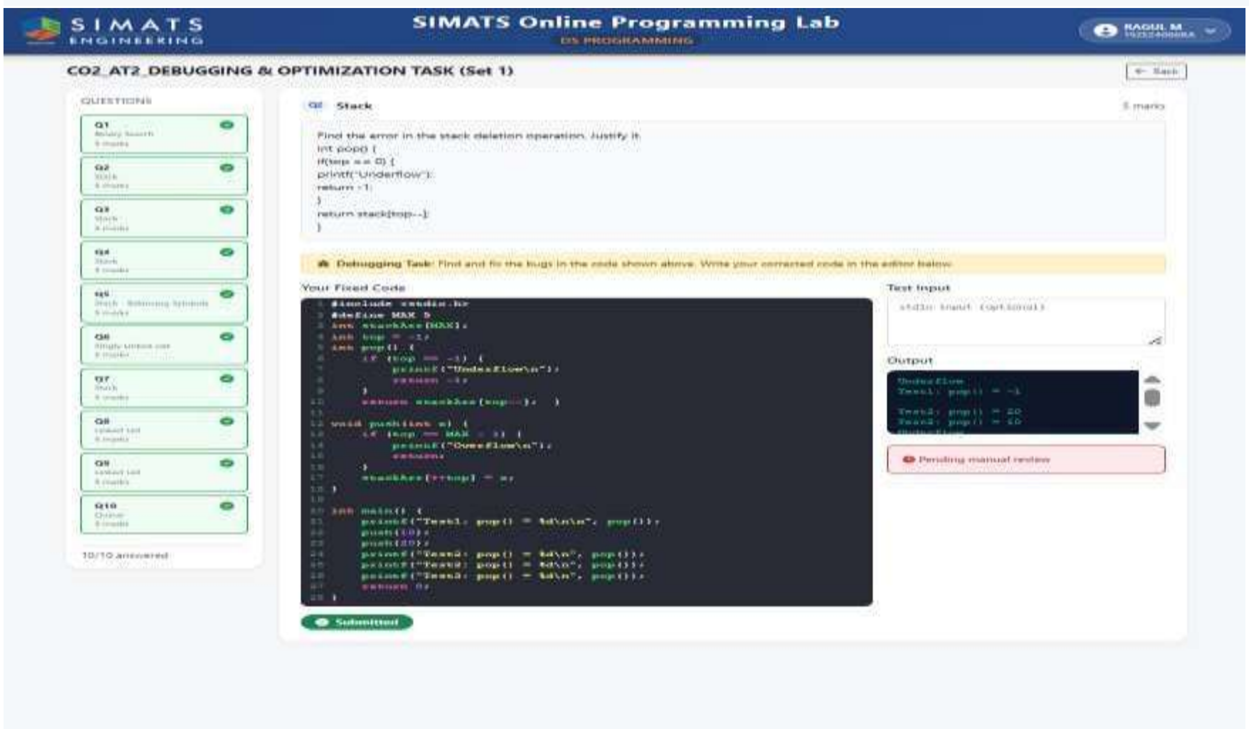
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
QUESTIONS
Binary Search
Your Fixed Code
Test Input
Output
Submitted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QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Stack
Find the error in the stack deletion operation. Justify it.
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#define MAX 5
Test Input
stdin input (optional)
Output
Pending manual review
Submitted

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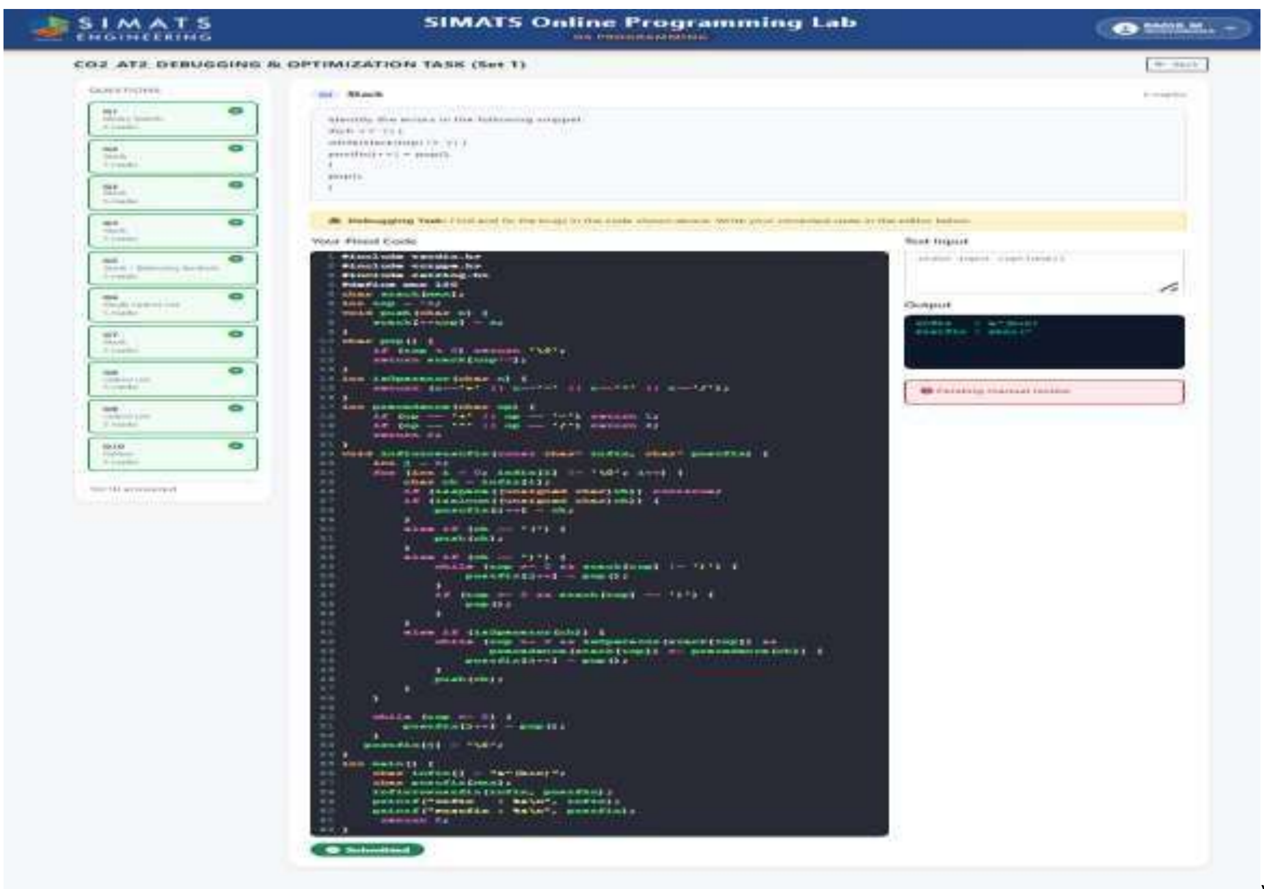
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Q1 Stack
5 marks
Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#include <stdio.h>
int stack[5];
int top;
void init() {
top = -1;
}
int main() {
init();
printf("top = %d\n", top);
return 0;
}
Test Input
stdin input (optional)
Output
top = -1
Pending manual review
Submitted



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Stack
Your Fixed Code
Test Input
Output
Submitted

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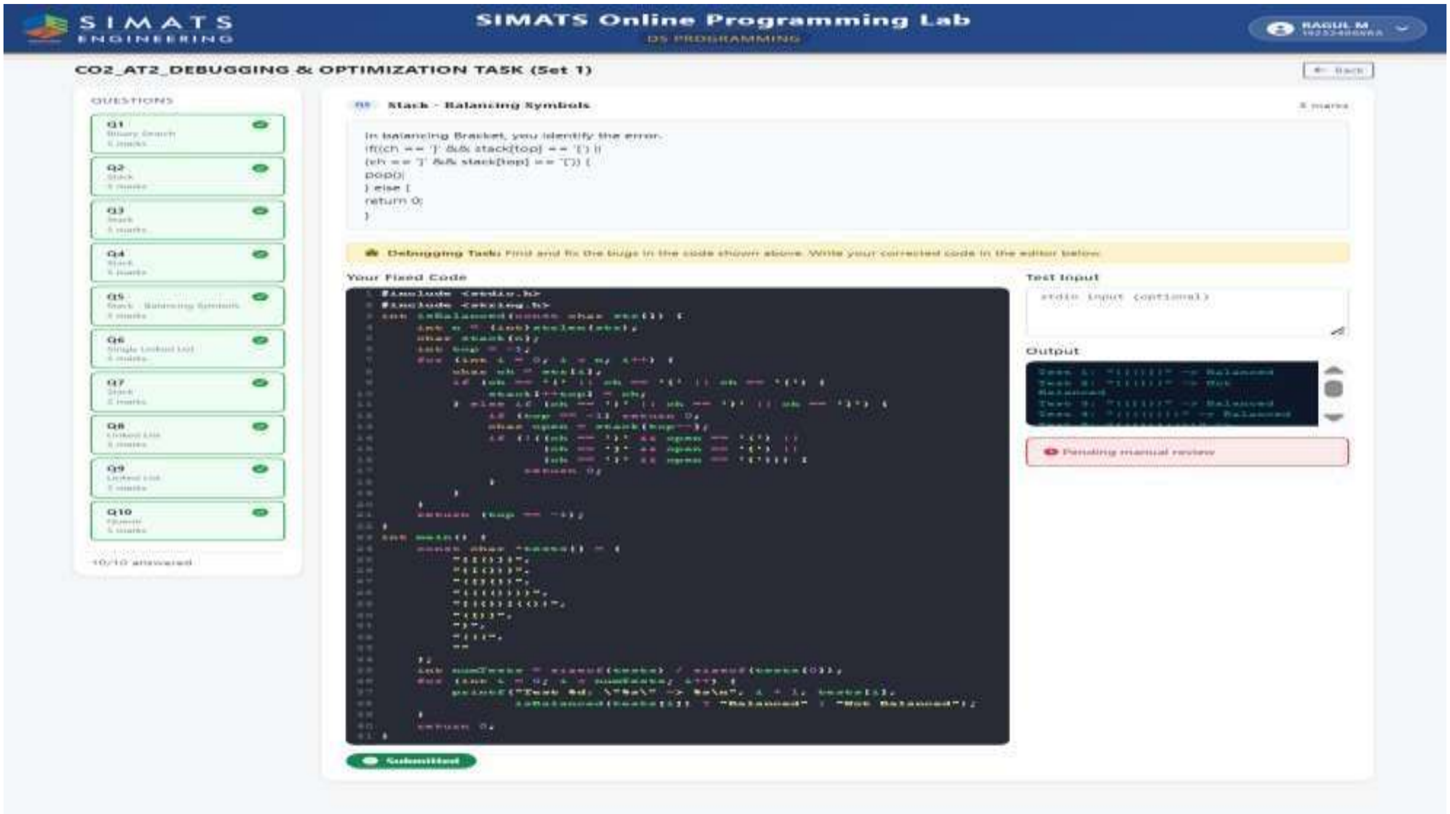
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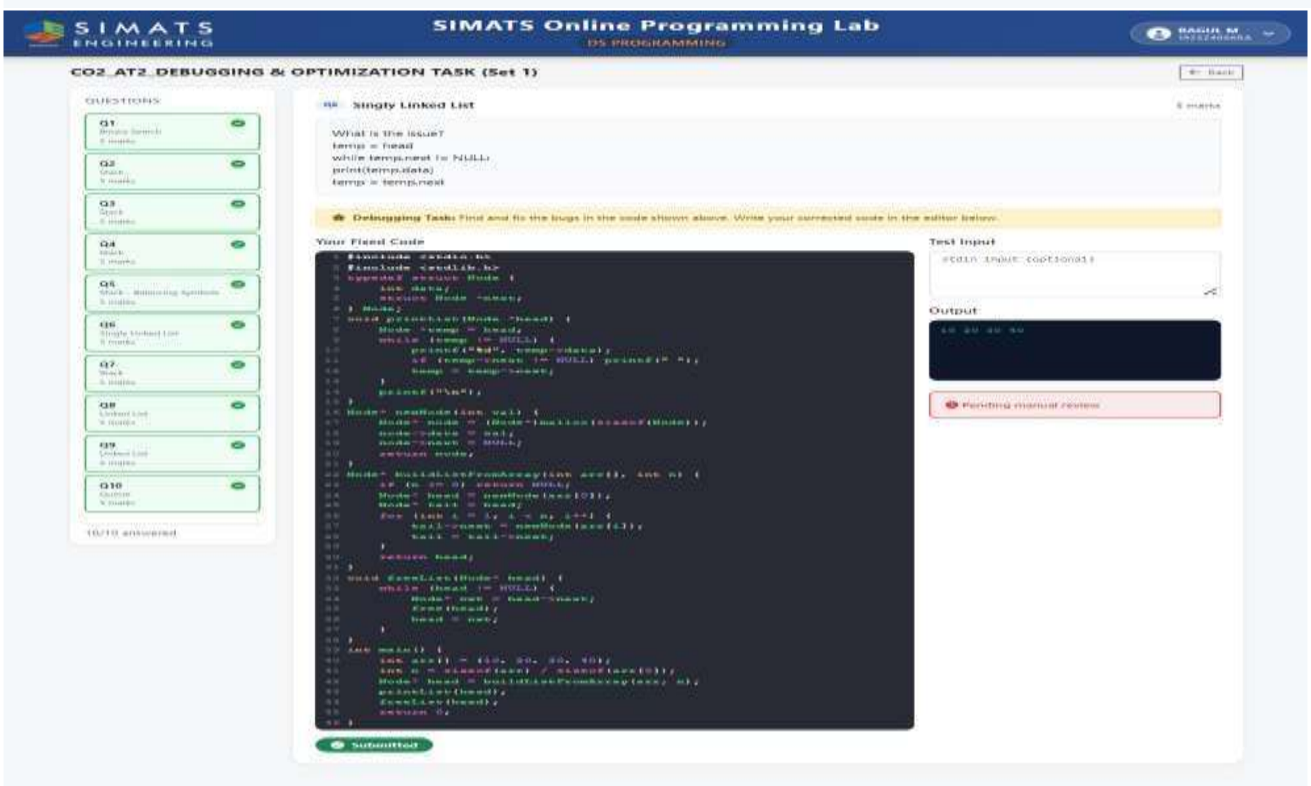
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Stack - Balancing Symbols
In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
pop();
} else {
return 0;
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#include <stdio.h>
#include <string.h>
Test Input
stdin input (optional)
Output
Pending manual review
Submitted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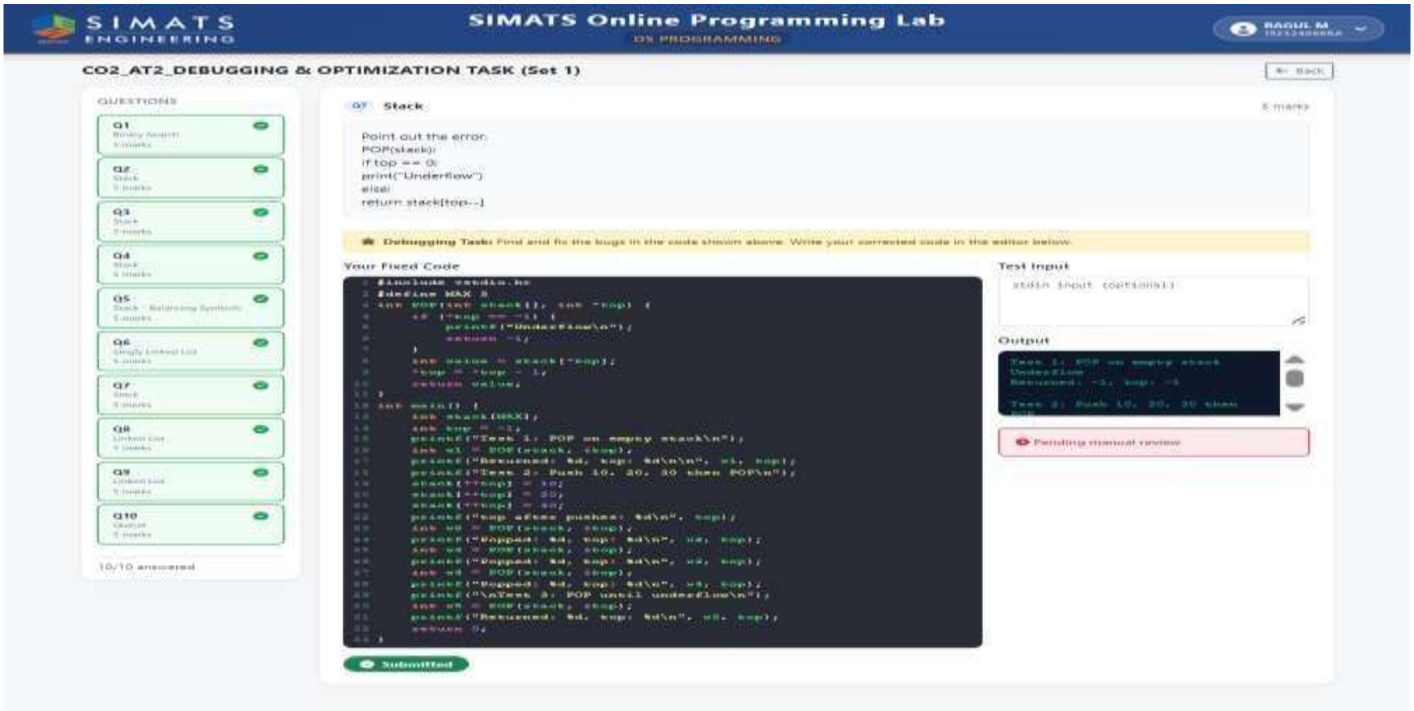
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Singly Linked List
What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#include <stdio.h>
#include <stdlib.h>
Test Input
stdin input (optional)
Output
Pending manual review
Submitted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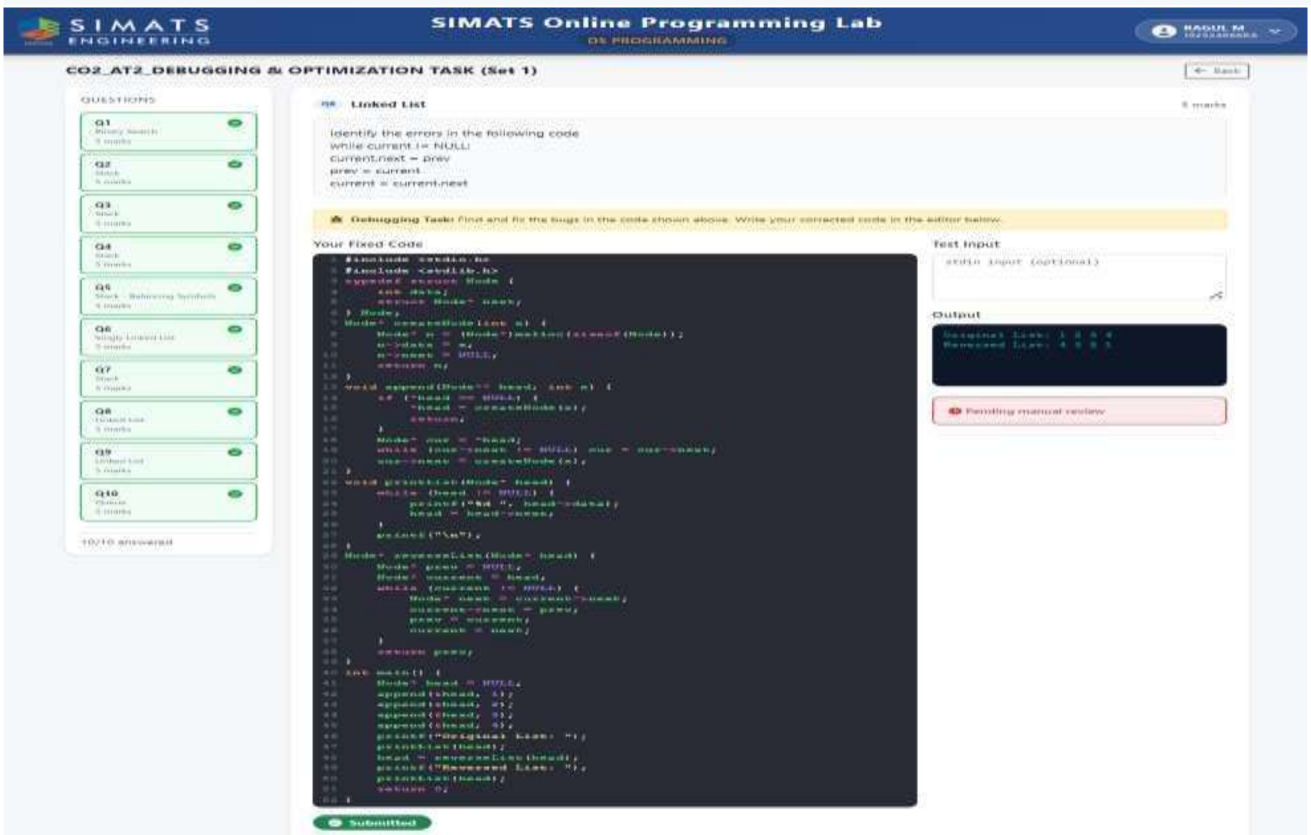
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Q7 Stack
Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]
Debugging Task:
Your Fixed Code
Test Input
Output
Submitted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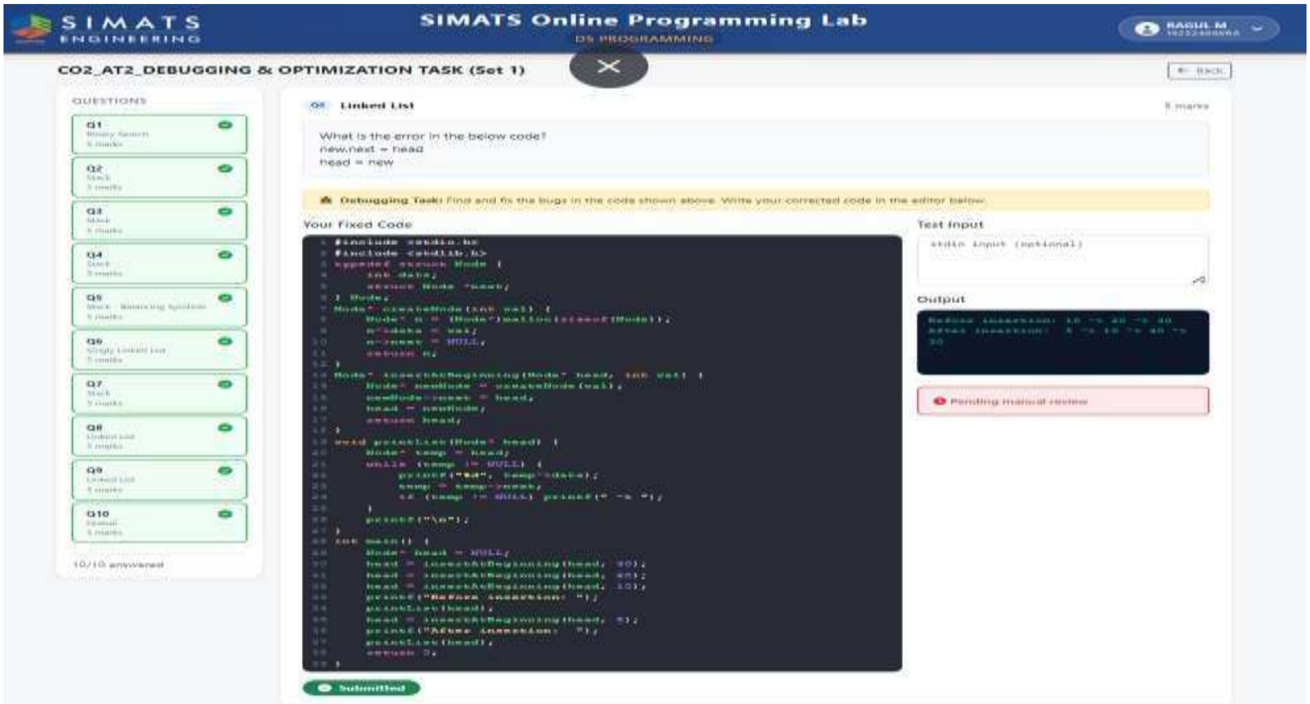
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Linked List
Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next
Debugging Task:
Your Fixed Code
Test Input
Output
Submitted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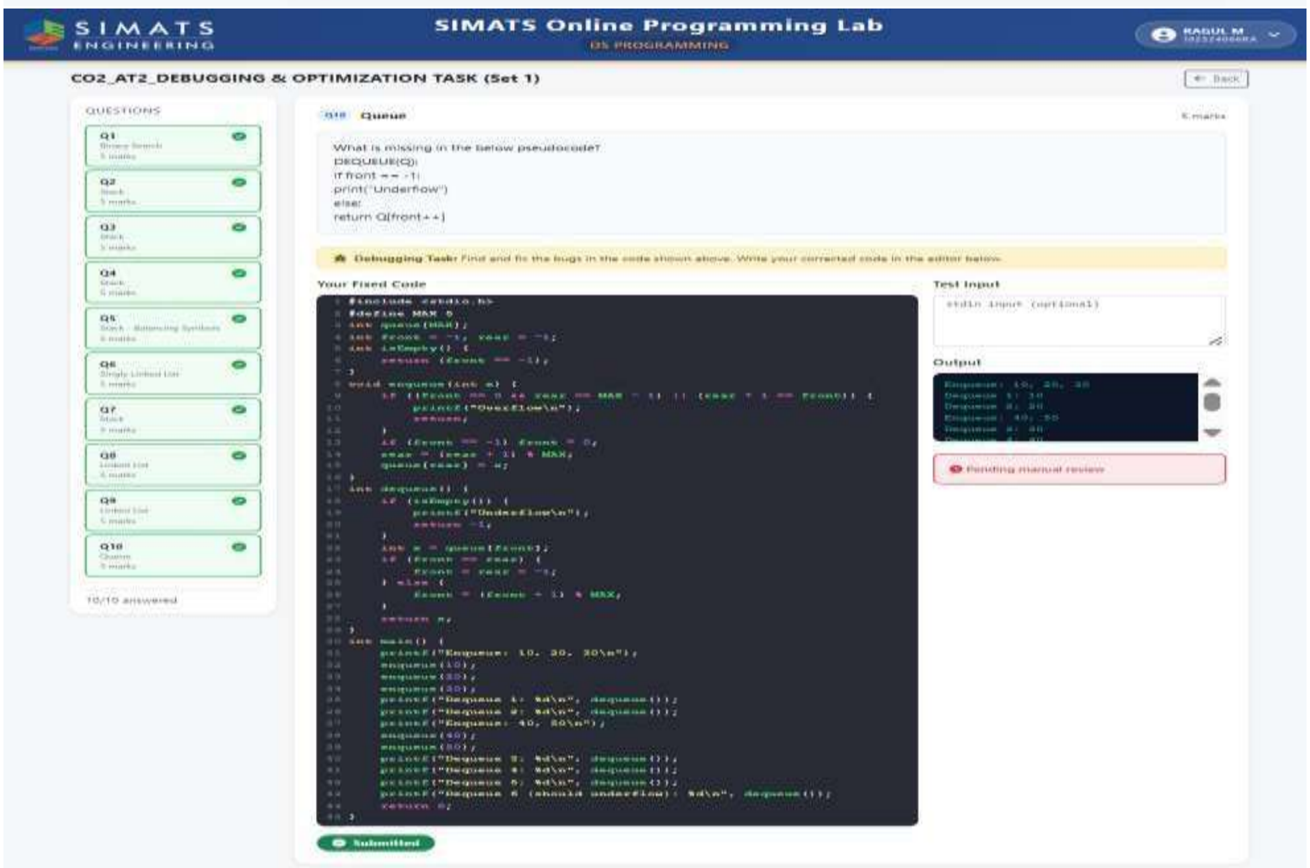
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Linked List
What is the error in the below code?
new.next = head
head = new
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
Test Input
stdin input (optional)
Output
Pending manual review
Submitted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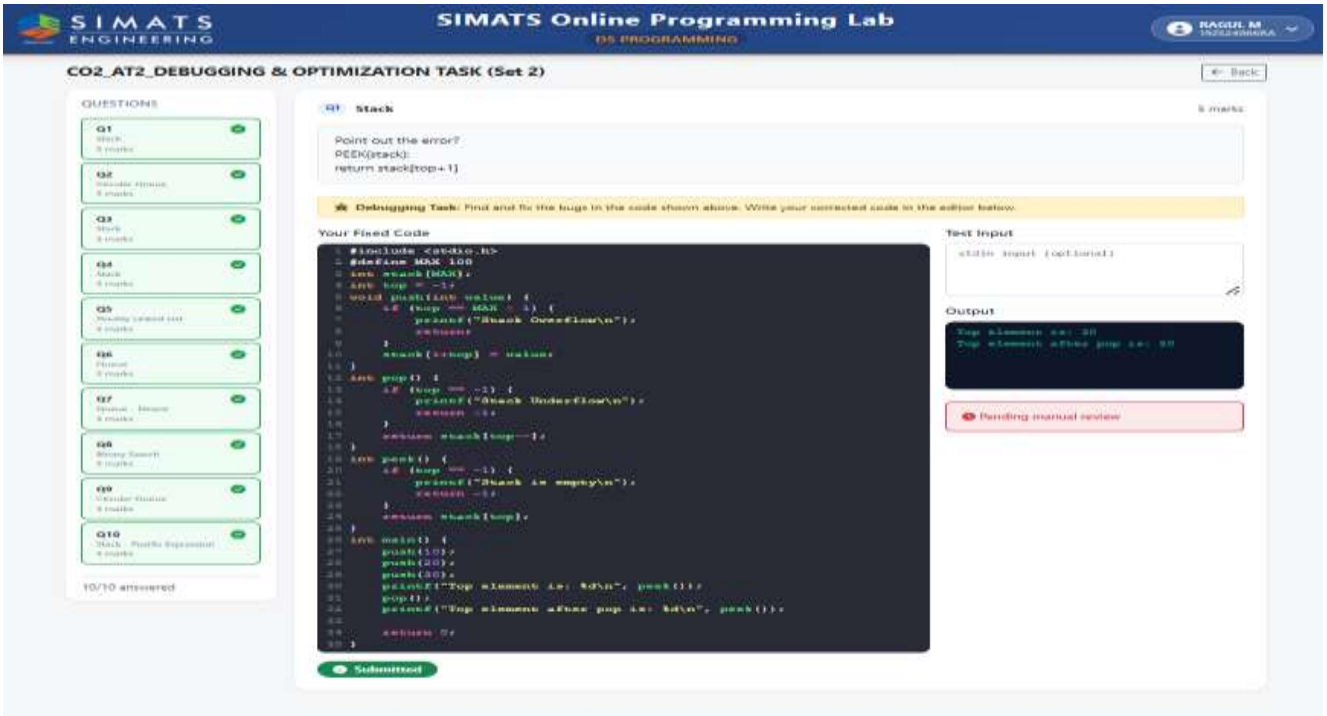
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Queue
What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
Test Input
stdin input (optional)
Output
Pending manual review
Submitted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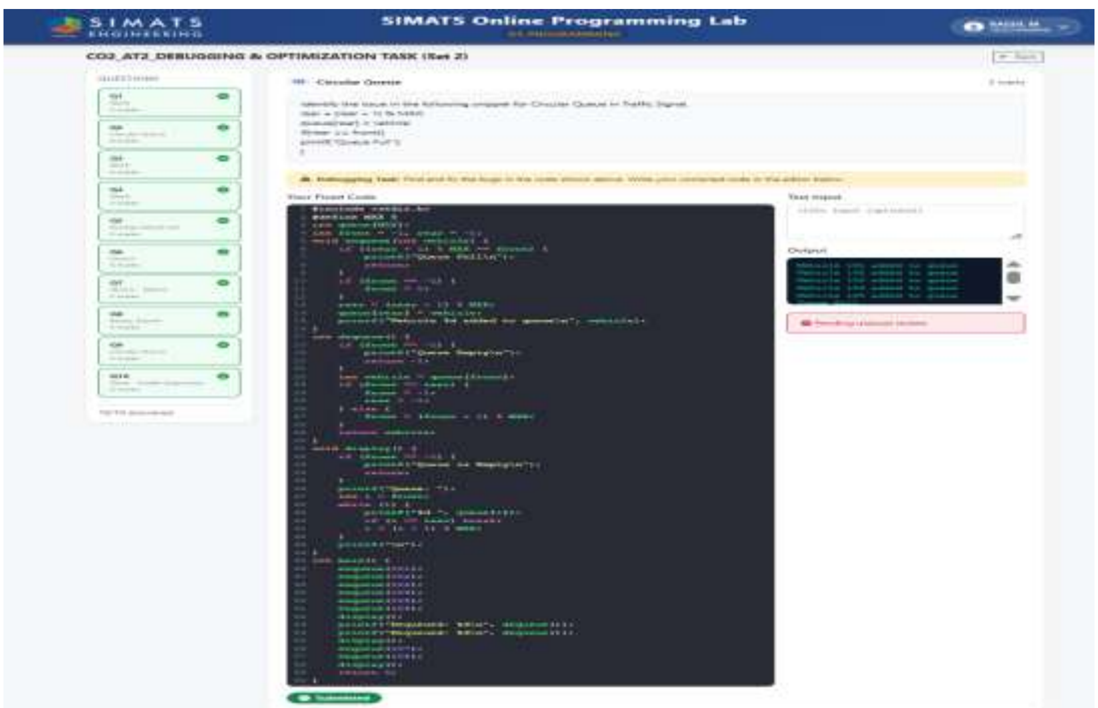
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Q1 Stack
5 marks
Point out the error?
PEEK(stack):
return stack[top+1]
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#include <stdio.h>
#define MAX 100
int stack[MAX];
int top = -1;
void push(int value) {
if (top == MAX - 1) {
printf("Stack Overflow\n");
return;
}
stack[++top] = value;
}
int pop() {
if (top == -1) {
printf("Stack Underflow\n");
return -1;
}
return stack[top--];
}
int peek() {
if (top == -1) {
printf("Stack is empty\n");
return -1;
}
return stack[top];
}
int main() {
push(10);
push(20);
push(30);
printf("Top element is: %d\n", peek());
pop();
printf("Top element after pop is: %d\n", peek());
return 0;
}
Submitted
Test Input
stdin input (optional)
Output
Top element is: 30
Top element after pop is: 20
Pending manual review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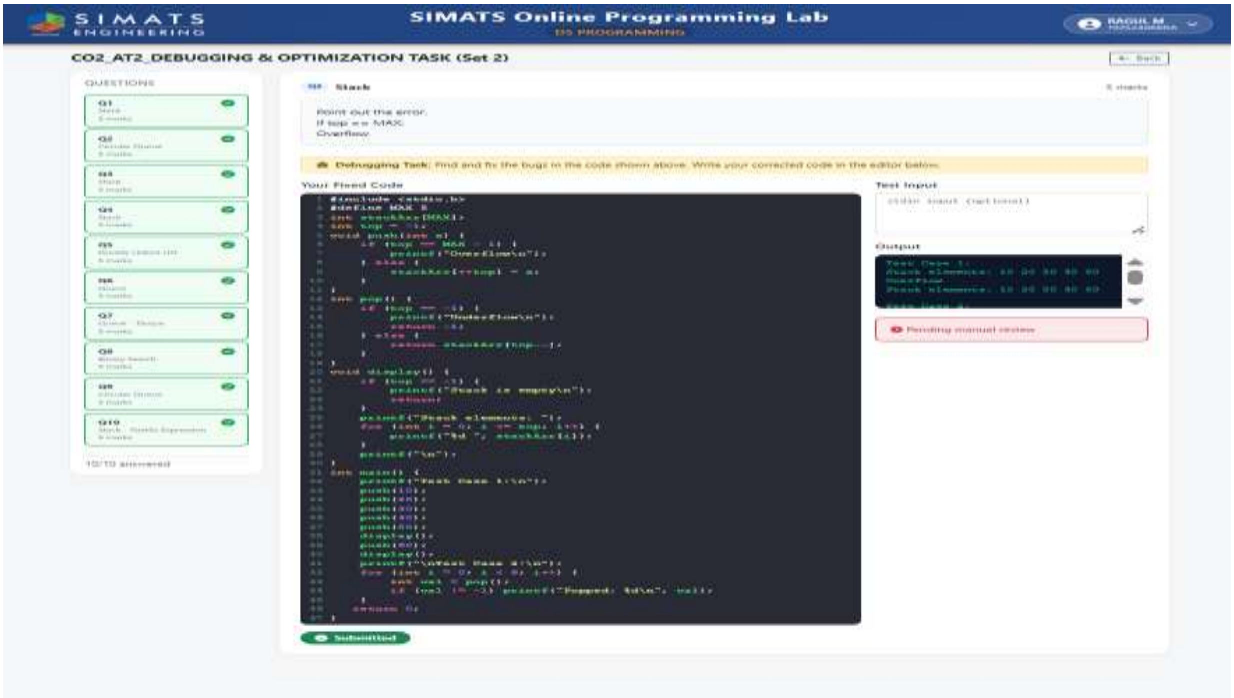
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
Circular Queue
Your Fixed Code
Test Input
Output
Submitted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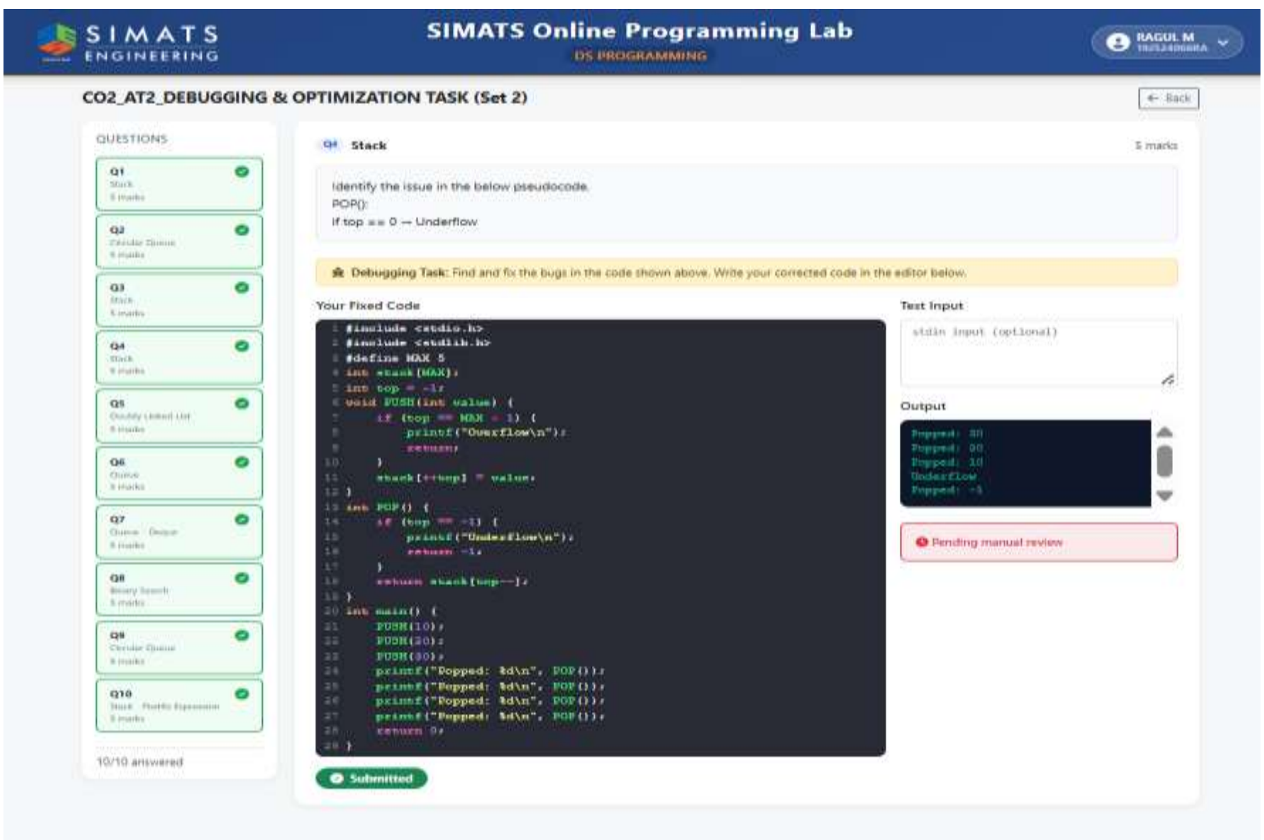
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
QUESTIONS
Stack
Point out the error.
If top == MAX
Overflow
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
Test Input
Output
Submitted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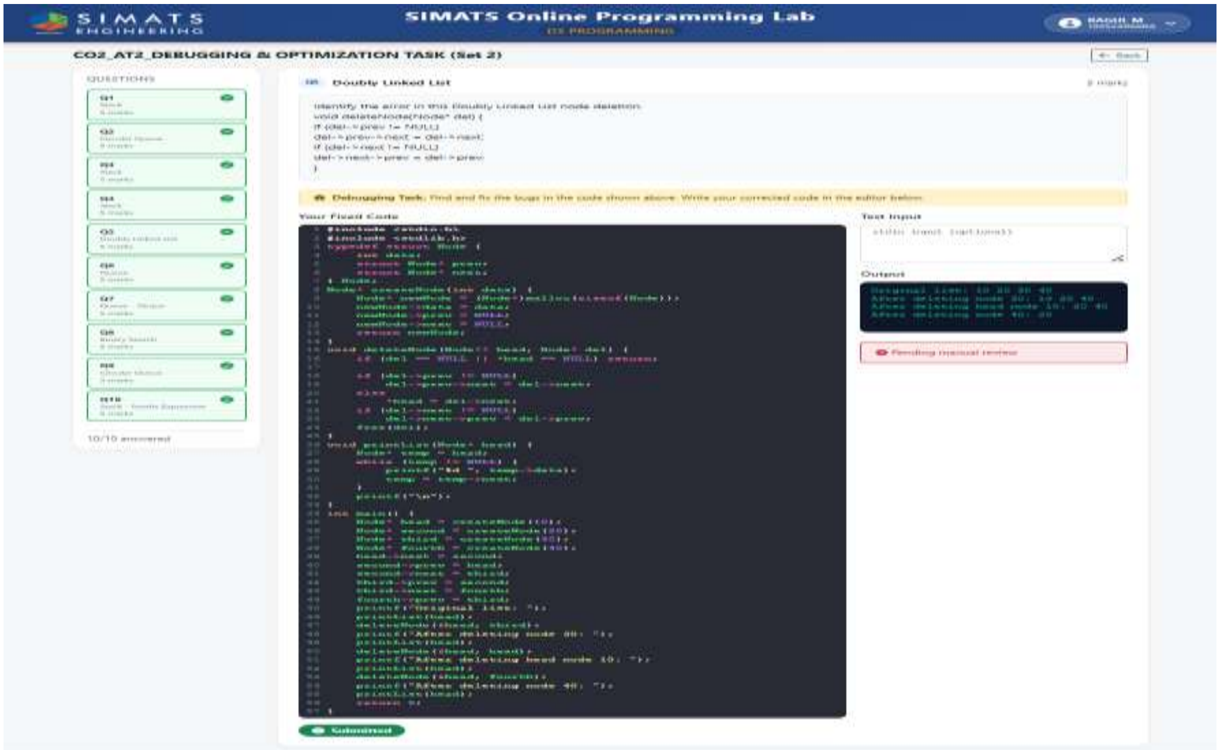
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
← Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
10/10 answered
Q4 Stack
5 marks
Identify the issue in the below pseudocode.
POP():
If top == 0 → Underflow
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
#include <stdio.h>
#include <stdlib.h>
#define MAX 5
int stack[MAX];
int top = -1;
void PUSH(int value) {
if (top == MAX - 1) {
printf("Overflow\n");
return;
}
stack[++top] = value;
}
int POP() {
if (top == -1) {
printf("Underflow\n");
return -1;
}
return stack[top--];
}
int main() {
PUSH(10);
PUSH(20);
PUSH(30);
printf("Popped: %d\n", POP());
printf("Popped: %d\n", POP());
printf("Popped: %d\n", POP());
printf("Popped: %d\n", POP());
return 0;
}
Test Input
stdin input (optional)
Output
Popped: 30
Popped: 20
Popped: 10
Underflow
Popped: -1
Pending manual review
Submitted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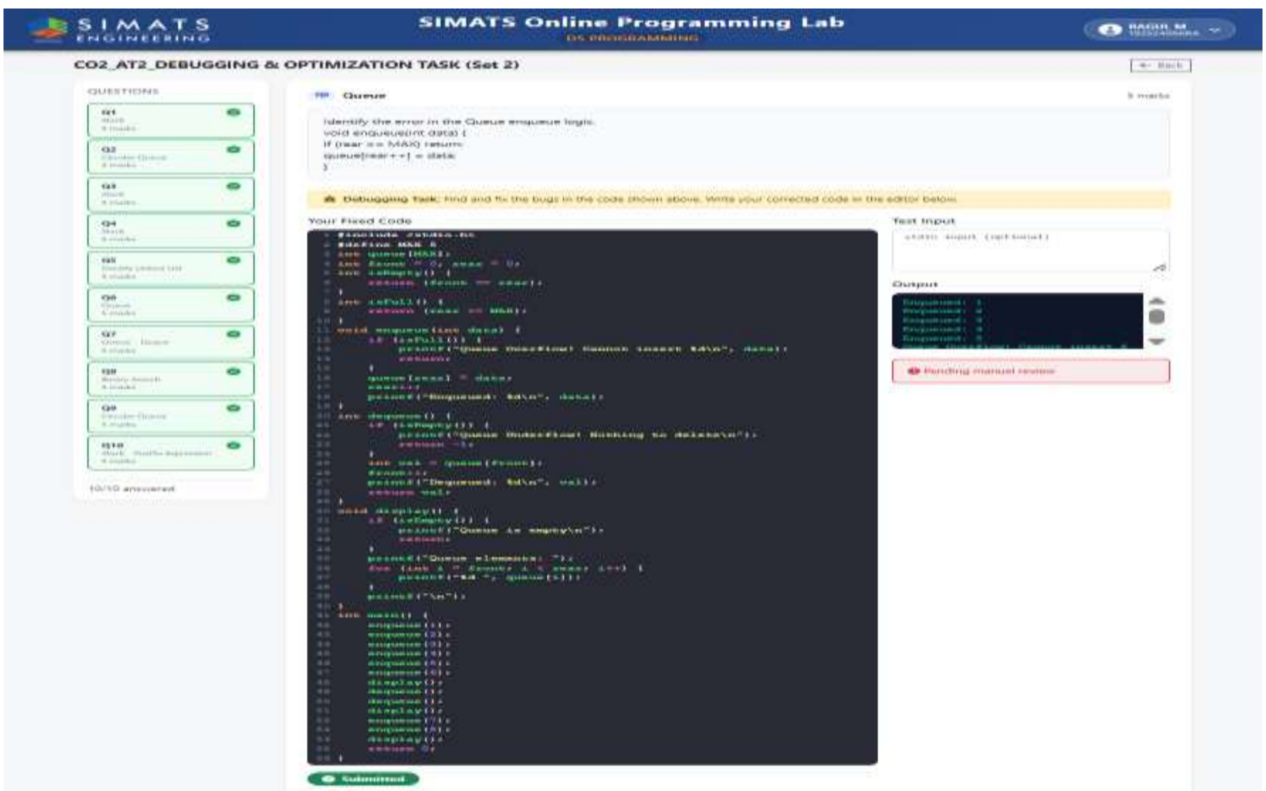
CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
QUESTIONS
Doubly Linked List
Debugging Task:
Your Fixed Code
Test Input
Output
10/10 answered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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
QUESTIONS
Queue
Identify the error in the Queue enqueue logic.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
Test Input
Output
10/10 answered

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 5
typedef struct {
    int deque[MAX];
    int front;
    int rear;
} Deque;
void initDeque(Deque *dq) {
    dq->front = -1;
    dq->rear = -1;
}
int isFull(Deque *dq) {
    return ((dq->front == 0 && dq->rear == MAX - 1) || (dq->front == d
}
int isEmpty(Deque *dq) {
    return (dq->front == -1);
}
void insertFront(Deque *dq, int data) {
    if (isFull(dq)) {
        printf("Overflow: Deque is full!\n");
        return;
    }
    if (dq->front == -1) {
        dq->front = 0;
        dq->rear = 0;
    } else if (dq->front == 0) {
        dq->front = MAX - 1;
    } else {
        dq->front = dq->front - 1;
    }
   dq->deque[dq->front] = data;
    printf("Inserted %d at front. Front: %d, Rear: %d\n", data, dq->fr
}
void insertRear(Deque *dq, int data) {
    if (isFull(dq)) {
        printf("Overflow: Deque is full!\n");
        return;
    }
    if (dq->front == -1) {
        dq->front = 0;
        dq->rear = 0;
    } else if (dq->rear == MAX - 1) {
        dq->rear = 0;
    } else {
        dq->rear = dq->rear + 1;
    }
    dq->deque[dq->rear] = data;
    printf("Inserted %d at rear. Front: %d, Rear: %d\n", data, dq->fro
}
void display(Deque *dq) {
    if (isEmpty(dq)) {
        printf("Deque is empty!\n");
        return;
    }
    printf("Deque elements: ");
    int i = dq->front;
    while (1) {
        printf("%d ", dq->deque[i]);
        if (i == dq->rear) break;
        i = (i + 1) % MAX;
    }
    printf("\n");
}
int main() {
    Deque dq;
    initDeque(&dq);
    printf("--- Testing Deque Operations ---\n");
    insertRear(&dq, 10);
    insertRear(&dq, 20);
    insertRear(&dq, 30);
    display(&dq);
    insertFront(&dq, 5);
    insertFront(&dq, 1);
    display(&dq);
    insertRear(&dq, 40);
    insertRear(&dq, 50);
    display(&dq);
   insertFront(&dq, 99);
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
--- Testing Deque Operations
---
Inserted 10 at rear. Front:
0, Rear: 0
Inserted 20 at rear. Front:
0, Rear: 1
```

Pending manual review

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) return -1;
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid - 1, x);
    return binarySearch(arr, mid + 1, r, x);
}
int main() {
    int arr[] = {1, 3, 5, 7, 9, 11, 13, 15};
    int n = sizeof(arr) / sizeof(arr[0]);
    int x = 11;
    int result = binarySearch(arr, 0, n - 1, x);
    if (result != -1)
        printf("Element %d found at index %d\n", x, result);
    else
        printf("Element %d not found\n", x);
    x = 8;
    result = binarySearch(arr, 0, n - 1, x);
    if (result != -1)
        printf("Element %d found at index %d\n", x, result);
    else
        printf("Element %d not found\n", x);
    return 0;
}
```

Test Input

```
stdin input (optional)
```

Output

```
Element 11 found at index 5
Element 8 not found
```

Pending manual review

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Dequeue 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic:

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#define max 5
int queue[max];
int front = -1, rear = -1;
int enqueue(int value) {
    if ((rear + 1) % max == front) {
        return 0;
    }
    if (front == -1) {
        front = rear = 0;
    } else {
        rear = (rear + 1) % max;
    }
    queue[rear] = value;
    return 1;
}
int dequeue() {
    if (front == -1) {
        return -1;
    }

    int data = queue[front];
    if (front == rear) {
        front = -1;
        rear = -1;
    } else {
        front = (front + 1) % max;
    }
    return data;
}
void printQueue() {
    if (front == -1) {
        printf("Queue is empty\n");
        return;
    }
    printf("Queue elements: ");
    int i = front;
    while (1) {
        printf("%d ", queue[i]);
        if (i == rear) break;
        i = (i + 1) % max;
    }
    printf("\n");
}
int main() {
    printf("Test 1: Dequeue from empty queue\n");
    printf("Dequeued: %d (expected -1)\n", dequeue());
    printQueue();
    printf("\n");
    printf("Test 2: Enqueue 10,20,30 then dequeue twice\n");
    enqueue(10);
    enqueue(20);
    enqueue(30);
    printQueue();
    printf("Dequeued: %d (expected 10)\n", dequeue());
    printf("Dequeued: %d (expected 20)\n", dequeue());
    printQueue();
    printf("\n");
    printf("Test 3: Make queue have one element and dequeue it\n");
    printf("Dequeued: %d (expected 30)\n", dequeue());
    printQueue();
    printf("Dequeued: %d (expected -1)\n", dequeue());
    printQueue();
    printf("\n");
    printf("Test 4: Circular wrap-around\n");
     enqueue(1);
    enqueue(2);
    enqueue(3);
    enqueue(4);
    printQueue();
    printf("Dequeued: %d (expected 1)\n", dequeue());
    printf("Dequeued: %d (expected 2)\n", dequeue());
    printQueue();
    enqueue(5);
    enqueue(6);
    printQueue();
   while (1) {
        int x = dequeue();
        if (x == -1) break;
        printf("Dequeued: %d\n", x);
    }
    printQueue();
   return 0;
}
```

Test Input

stdin input (optional)

Output

```
Test 1: Dequeue from empty
queue
Dequeued: -1 (expected -1)
Queue is empty

Test 2: Enqueue 10,20,30 then
```

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SIMATS ENGINEERING

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DS PROGRAMMING

RAGUL M

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Dequeue 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
#define MAX 1000
typedef struct {
    int a[MAX];
    int top;
} Stack;
void init(Stack *s) {
    s->top = -1;
}
int isEmpty(Stack *s) {
    return s->top == -1;
}
void push(Stack *s, int x) {
    if (s->top >= MAX - 1) return;
    s->a[++s->top] = x;
}
int pop(Stack *s) {
    if (isEmpty(s)) return 0;
    return s->a[s->top--];
}
int evalPostfix(const char *expr) {
    Stack s;
    init(&s);
    int i = 0;
    while (expr[i] != '\0') {
        if (expr[i] == ' ' || expr[i] == '\n' || expr[i] == '\t') {
            i++;
            continue;
        }
        if (isdigit((unsigned char)expr[i])) {
            int num = 0;
            while (isdigit((unsigned char)expr[i])) {
                num = num * 10 + (expr[i] - '0');
                i++;
            }
            push(&s, num);
            continue;
        }
        char op = expr[i];
        int op1 = pop(&s);
        int op2 = pop(&s);
        int result = 0;

        switch (op) {
            case '+': result = op2 + op1; break;
            case '-': result = op2 - op1; break;
            case '*': result = op2 * op1; break;
            case '/': result = op2 / op1; break;
            default:  result = 0; break;
        }

        push(&s, result);
        i++;
    }

    return pop(&s);
}

int main() {
    char expr[2000];
    const char *tests[] = {
        "2 3 +",
        "5 1 2 + 4 * + 3 -",
        "10 2 /",
        "7 3 -",
        "12 3 * 4 +"
    };

    for (int t = 0; t < 5; t++) {
        strcpy(expr, tests[t]);
        printf("Postfix: %s\n", expr);
        printf("Result : %d\n\n", evalPostfix(expr));
    }

    return 0;
}
```

Submitted

Test Input

stdin input (optional)

Output

```
Postfix: 2 3 +
Result : 5

Postfix: 5 1 2 + 4 * + 3 -
Result : 14
```

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